EXPERIMENT NO. 15

IMPLEMENT A SIMPLE REINFORCEMENT ALGORITHM FOR AN NLP PROBLEM

DATE: _____

AIM

To implement a simple reinforcement learning algorithm for an NLP problem using Q-learning to select the best response for a given user input.

ALGORITHM

1. Define the states representing user inputs.
2. Define the possible responses as actions.
3. Initialize the Q-table with zeros.
4. Select an action using an epsilon-greedy policy.
5. Give a reward for a correct response and a penalty for an incorrect response.
6. Update the Q-table using the Q-learning formula.
7. Display the best response for each user input.

PROGRAM

```
import numpy as np
import random

# States: different user inputs
states = ["hello", "thanks", "bye"]

# Actions: possible chatbot responses
actions = ["greeting", "welcome", "goodbye"]

# Correct response for each state
correct_action = [0, 1, 2]

# Initialize Q-table
Q = np.zeros((len(states), len(actions)))

# Q-learning parameters
alpha = 0.1
gamma = 0.9
epsilon = 0.2

# Training
for episode in range(1000):
    for state in range(len(states)):

        # Epsilon-greedy action selection
        if random.random() < epsilon:
            action = random.randint(0, len(actions) - 1)
        else:
            action = np.argmax(Q[state])

        # Reward for the selected response
        if action == correct_action[state]:
            reward = 1
        else:
            reward = -1

        # Q-learning update (same state as next state)
        Q[state, action] += alpha * (
            reward + gamma * np.max(Q[state]) - Q[state, action]
        )
```

```
# Display best responses
print("Best chatbot responses:")
for i, state in enumerate(states):
    best_action = np.argmax(Q[i])
    print(state, "->", actions[best_action])
```

OUTPUT

```
Best chatbot responses:\nhello -> greeting\nthanks -> welcome\nbye -> goodbye
```

RESULT

Thus, the simple Q-learning reinforcement algorithm to select suitable chatbot responses for different user inputs has been implemented successfully and verified.

EXECUTION: Install NumPy using `pip install numpy`, save the program as `experiment15.py`, and run it using `python experiment15.py`.

Note: This is a small demonstration of response selection with predefined input states and rewards. It is not a full conversational NLP model.